

- **Attention deficit hyperactivity disorder (ADHD) (Kapvay, ages 6 to 17)**
- Attention deficit hyperactivity disorder (ADHD)
- Tourette's syndrome
- Substance withdrawal, including opiates and alcohol
- Anxiety disorders, including posttraumatic stress disorder (PTSD) and social anxiety disorder
- Clozapine-induced hypersalivation
- Menopausal flushing
- Severe pain in cancer patients that is not adequately relieved by opioid analgesics alone (combination with opiates)

How the Drug Works

- For ADHD, theoretically has central actions on postsynaptic alpha 2 receptors in the prefrontal cortex
- For hypertension, stimulates alpha 2 adrenergic receptors in the brain stem, reducing sympathetic outflow from the CNS and decreasing peripheral resistance, renal vascular resistance, heart rate, and blood pressure
- An imidazoline, so also interacts at imidazoline receptors

How Long Until It Works

- For ADHD, can take a few weeks to see maximum therapeutic benefits
- Blood pressure may be lowered 30–60 minutes after first dose; greatest reduction seen after 2–4 hours
- May take several weeks to control blood pressure adequately

If It Works

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning
- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Reevaluate the need for treatment periodically
- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented
- For hypertension, continue treatment indefinitely and check blood pressure regularly

If It Doesn't Work

- Consider adjusting dose or switching to another agent
- Consider behavioral therapy

- Consider the presence of noncompliance and counsel patient and parents
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness, etc.)

Best Augmenting Combos for Partial Response or Treatment Resistance

- Best to attempt another monotherapy prior to augmenting for ADHD
- Possibly combination with stimulants (with caution as benefits of combination poorly documented and there are some reports of serious adverse events)
- Combinations for ADHD should be for the expert, while monitoring the patient closely, and when other treatment options have failed
- Chlorthalidone, thiazide-type diuretics, and furosemide for hypertension

Tests

- Blood pressure should be checked regularly during treatment

Side Effects

How Drug Causes Side Effects

- Excessive actions on alpha 2 receptors and/or on imidazoline receptors

Notable Side Effects

- * Dry mouth
- * Dizziness, constipation, sedation
- Weakness, fatigue, impotence, loss of libido, insomnia, headache
- Major depression
- Dermatologic reactions (especially with transdermal clonidine)
- Hypotension, occasional syncope
- Tachycardia
- Nervousness, agitation
- Nausea, vomiting

Life-Threatening or Dangerous Side Effects

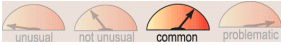
- Sinus bradycardia, atrioventricular block
- During withdrawal, hypertensive encephalopathy, cerebrovascular accidents, and death (rare)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Some patients may not tolerate it
- Can abate with time

What to Do About Side Effects

- Wait
- Take larger dose at bedtime to avoid daytime sedation
- Switch to another medication with better evidence of efficacy
- * For withdrawal and discontinuation reactions, may need to reinstate clonidine and taper very slowly when stabilized

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Extended-release for ADHD: 0.1–0.4 mg/day in divided doses

- Immediate-release for hypertension: 0.2–0.6 mg/day in divided doses
- Opioid withdrawal: 0.1 mg 3 times daily (can be higher in inpatient setting)

Dosage Forms

- Extended-release tablet 0.1 mg, 0.2 mg
- Immediate-release tablet 0.1 mg scored, 0.2 mg scored, 0.3 mg scored
- Topical (7 day administration) 0.1 mg/24 hours, 0.2 mg/24 hours, 0.3 mg/24 hours
- Injection 0.1 mg/mL, 0.5 mg/mL

How to Dose

- Oral (for ADHD): initial 0.1 mg at bedtime; can increase by 0.1 mg/day each week with dosing divided and larger dose at bedtime; maximum dose generally 0.4 mg/day in divided doses
- For opioid withdrawal: 0.1 mg 3 times daily; next dose should be withheld if blood pressure falls below 90/60 mmHg; outpatients should not be given more than a 3-day supply, detoxification can usually be achieved in 4–6 days for short-acting opioids
- Oral (for hypertension): initial 0.1 mg in 2 divided doses, morning and night; can increase by 0.1 mg/day each week; maximum dose generally 2.4 mg/day

- Topical (for hypertension): apply once every 7 days in hairless area; change location with each application
- Injection (for hypertension): initial 30 μ g/hour; maximum 40 μ g/hour; 500 mg/mL must be diluted



Dosing Tips

- Extended-release tablet should not be chewed, crushed, or broken before swallowing, as this could alter controlled-release properties
- Do not substitute different clonidine products for each other on a mg-per-mg basis, because they have different pharmacokinetic profiles
- Adverse effects are dose-related and usually transient
- The last dose of the day should occur at bedtime so that blood pressure is controlled overnight
- If clonidine is terminated abruptly, rebound hypertension may occur within 2–4 days, so taper dose in decrements of no more than 0.1 mg every 3 to 7 days when discontinuing
- Using clonidine in combination with another antihypertensive agent may attenuate the development of tolerance to clonidine's antihypertensive effects
- The likelihood of severe discontinuation reactions with CNS and cardiovascular symptoms may be greater after administration of high doses of clonidine

- * In patients who have developed localized contact sensitization to transdermal clonidine, continuing transdermal dosing on other skin areas or substituting with oral clonidine may be associated with the development of a generalized skin rash, urticaria, or angioedema
- * If administered with a beta blocker, stop the beta blocker first for several days before the gradual discontinuation of clonidine in cases of planned discontinuation

Overdose

- Hypotension, hypertension, miosis, respiratory depression, seizures, bradycardia, hypothermia, coma, sedation, decreased reflexes, weakness, irritability, dysrhythmia

Long-Term Use

- Patients may develop tolerance to the antihypertensive effects
- * Studies have not established the utility of clonidine for long-term CNS uses
- * Be aware that forgetting to take clonidine or running out of medication can lead to abrupt discontinuation and associated withdrawal reactions and complications

Habit Forming

- Reports of some abuse by opiate addicts
- Reports of some abuse by non-opioid-dependent patients

How to Stop

- * Discontinuation reactions are common and sometimes severe
- Sudden discontinuation can result in nervousness, agitation, headache, and tremor, with rapid rise in blood pressure
- Rare instances of hypertensive encephalopathy, cerebrovascular accident, and death have been reported after clonidine withdrawal
- Taper over 2–4 days or longer to avoid rebound effects (nervousness, increased blood pressure)
- If administered with a beta blocker, stop the beta blocker first for several days before the gradual discontinuation of clonidine

Pharmacokinetics

- Half-life 12–16 hours
- Metabolized by the liver
- Excreted renally



Drug Interactions

- The likelihood of severe discontinuation reactions with CNS and cardiovascular symptoms may be greater when clonidine is combined with beta blocker treatment
- Increased depressive and sedative effects when taken with other CNS depressants
- TCAs may reduce the hypotensive effects of clonidine

- Corneal lesions in rats increased by use of clonidine with amitriptyline
- Use of clonidine with agents that affect sinus node function or AV nodal function (e.g., digitalis, calcium channel blockers, beta blockers) may result in bradycardia or AV block

Other Warnings/Precautions

- There have been cases of hypertensive encephalopathy, cerebrovascular accidents, and death after abrupt discontinuation
- If used with a beta blocker, the beta blocker should be stopped several days before tapering clonidine
- In patients who have developed localized contact sensitization to transdermal clonidine, continuing transdermal dosing on other skin areas or substituting with oral clonidine may be associated with the development of a generalized skin rash, urticaria, or angioedema
- Use with caution in patients at risk for hypotension, heart block, and bradycardia
- Be aware that forgetting to take clonidine or running out of medication can lead to abrupt discontinuation and associated withdrawal reactions and complications
- Injection is not recommended for use in managing obstetrical, postpartum, or peri-operative pain
- Certain transdermal patches containing even small traces of aluminum or other metals in the adhesive backing can cause skin

burns if worn during MRI, so warn patients taking the transdermal formulation about this possibility and advise them to disclose this information if they need an MRI

Do Not Use

- If there is a proven allergy to clonidine

Special Populations

Renal Impairment

- Use with caution and possibly reduce dose

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Use with caution in patients with recent myocardial infarction, severe coronary insufficiency, cerebrovascular disease

Elderly

- Elderly patients may tolerate a lower initial dose better
- Elderly patients may be more sensitive to sedative effects



Children and Adolescents

- Safety and efficacy not established for children under age 6
- Children may be more sensitive to hypertensive effects of withdrawing treatment
- * Because children commonly have gastrointestinal illnesses that lead to vomiting, they may be more likely to abruptly discontinue clonidine and therefore be more susceptible to hypertensive episodes resulting from abrupt inability to take medication
- Children may be more likely to experience CNS depression with overdose and may even exhibit signs of toxicity with 0.1 mg of clonidine
- Injection may be used in pediatric cancer patients with severe pain unresponsive to other medications

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Some animal studies have shown adverse effects
- Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus

* For ADHD patients, clonidine should generally be discontinued before anticipated pregnancies

Breast Feeding

- Some drug is found in mother's breast milk
- No adverse effects have been reported in nursing infants
- If irritability or sedation develop in nursing infant, may need to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Not a controlled substance

Potential Disadvantages

- Not well studied in adults with ADHD
- Withdrawal reactions
- Noncompliant patients
- Patients on concomitant CNS medications

Primary Target Symptoms

- Concentration
- Motor hyperactivity

- Oppositional and impulsive behavior
- High blood pressure

Pearls

* Clonidine extended-release is approved for ADHD in children ages 6–17

- As monotherapy or in combination with methylphenidate for ADHD with conduct disorder or oppositional defiant disorder, may improve aggression, oppositional, and conduct disorder symptoms
- Clonidine is sometimes used in combination with stimulants to reduce side effects and enhance therapeutic effects on motor hyperactivity
- Doses of 0.1 mg in 3 divided doses have been reported to reduce stimulant-induced insomnia as well as impulsivity

* Clonidine may also be effective for treatment of tic disorders, including Tourette's syndrome

- May suppress tics especially in severe Tourette's syndrome, and may be even better at reducing explosive violent behaviors in Tourette's syndrome
- Sedation is often unacceptable in various patients despite improvement in CNS symptoms and leads to discontinuation of treatment, especially for ADHD and Tourette's syndrome
- Considered an investigational treatment for most other CNS applications

- May block the autonomic symptoms in anxiety and panic disorders (e.g., palpitations, sweating) and improve subjective anxiety as well
- May be useful in decreasing the autonomic arousal of PTSD
- May be useful as an as-needed medication for stage fright or other predictable socially phobic situations
- May also be useful when added to SSRIs for reducing arousal and dissociative symptoms in PTSD
- May block autonomic symptoms of opioid withdrawal (e.g., palpitations, sweating) especially in inpatients, but muscle aches, irritability, and insomnia may not be well suppressed by clonidine
- Often prescribed with naltrexone to suppress symptoms of opioid withdrawal; this requires monitoring of the patient for 8 hours on the first day due to the potential severity of naltrexone-induced withdrawal and the potential blood pressure effects of clonidine
- May be useful in decreasing the hypertension, tachycardia, and tremulousness associated with alcohol withdrawal, but not the seizures or delirium tremens in complicated alcohol withdrawal
- Clonidine may improve social relationships, affectual responses, and sensory responses in autistic disorder
- Clonidine may reduce the incidence of menopausal flushing
- Growth hormone response to clonidine may be reduced during menses

- Clonidine stimulates growth hormone secretion (no chronic effects have been observed)
- Alcohol may reduce the effects of clonidine on growth hormone
- * Guanfacine is a related centrally active alpha 2 agonist hypotensive agent that has been used for similar CNS applications but has not been as widely investigated or used as clonidine
- * Guanfacine may be tolerated better than clonidine in some patients (e.g., sedation) or it may work better in some patients for CNS applications than clonidine, but no head-to-head trials

Suggested Reading

American Psychiatric Association . Practice guideline for the treatment of patients with substance use disorders, second edition . *Am J Psychiatry* 2007 ;**164** (4):1 –86 .

Burris JF . The USA experience with the clonidine transdermal therapeutic system . *Clin Auton Res* 1993 ;**3** :391–6 .

Croxtall JD. Clonidine extended-release: in attention-deficit hyperactivity disorder . *Paediatr Drugs* 2011 ;**13** (5):329–36 .

Gavras I , Manolis AJ , Gavras H . The alpha2-adrenergic receptors in hypertension and heart failure: experimental and clinical studies . *J Hypertens* 2001 ;**19** :2115–24 .

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Neil MJ. Clonidine: clinical pharmacology and therapeutic use in pain management . *Curr Clin Pharmacol* 2011 ;**6** (4):280–7 .

Clorazepate

Therapeutics

Brands

- Azene
- Tranxene
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (anxiolytic)

Commonly Prescribed for

(bold for FDA approved)

- **Anxiety disorder**
- **Symptoms of anxiety**
- **Acute alcohol withdrawal**
- Partial seizures (adjunct)

- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibits neuronal activity presumably in amygdala-centered fear circuits to provide therapeutic benefits in anxiety disorders

How Long Until It Works

- Some immediate relief with first dosing is common; can take several weeks with daily dosing for maximal therapeutic benefit

If It Works

- For short-term symptoms of anxiety –after a few weeks, discontinue use or use on an “as-needed” basis
- For chronic anxiety disorders, the goal of treatment is complete remission of symptoms as well as prevention of future relapses
- For chronic anxiety disorders, treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- For long-term symptoms of anxiety, consider switching to an SSRI or SNRI for long-term maintenance

- If long-term maintenance with a benzodiazepine is necessary, continue treatment for 6 months after symptoms resolve, and then taper dose slowly
- If symptoms reemerge, consider treatment with an SSRI or SNRI, or consider restarting the benzodiazepine; sometimes benzodiazepines have to be used in combination with SSRIs or SNRIs for best results

If It Doesn't Work

- Consider switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy, especially cognitive behavioral psychotherapy
- Consider presence of concomitant substance abuse
- Consider presence of clonazepam abuse
- Consider another diagnosis, such as a comorbid medical condition

Best Augmenting Combos for Partial Response or Treatment Resistance

- Benzodiazepines are frequently used as augmenting agents for antipsychotics and mood stabilizers in the treatment of psychotic and bipolar disorders
- Benzodiazepines are frequently used as augmenting agents for SSRIs and SNRIs in the treatment of anxiety disorders
- Not generally rational to combine with other benzodiazepines

- Caution if using as an anxiolytic concomitantly with other sedative hypnotics for sleep

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

-  Sedation, fatigue, depression
 -  Dizziness, ataxia, slurred speech, weakness
 -  Forgetfulness, confusion
 -  Nervousness
- Rare hallucinations, mania

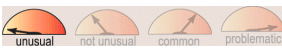
- Rare hypotension
- Hypersalivation, dry mouth



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Especially at initiation of treatment or when dose increases
- Tolerance often develops over time

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose

- Take largest dose at bedtime to avoid sedative effects during the day
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Anxiety: 15–60 mg/day in divided doses
- Alcohol withdrawal: 30–60 mg/day in divided doses

Dosage Forms

- Tablet 3.75 mg scored, 7.5 mg scored, 15 mg scored

How to Dose

- Anxiety: initial 15 mg/day in divided doses; adjust dose as needed on subsequent days; single-dose tablet may be given once daily at bedtime after patient is stable; maximum generally 90 mg/day
- Alcohol withdrawal: initial 30 mg, then 30–60 mg in divided doses; second day 45–90 mg in divided doses; third day 22.5–45 mg in divided doses; fourth day 15–30 mg in divided doses; after fourth

day decrease dose gradually and discontinue when patient is stable;
maximum generally 90 mg/day

- Epilepsy: initial 7.5 mg 3 times/day; increase by 7.5 mg weekly;
maximum generally 90 mg/day



Dosing Tips

- Use lowest possible effective dose for the shortest possible period of time (a benzodiazepine-sparing strategy)
- Assess need for continued treatment regularly
- Risk of dependence may increase with dose and duration of treatment
- For interdose symptoms of anxiety, can either increase dose or maintain same total daily dose but divide into more frequent doses
- Can also use an as-needed occasional “top-up” dose for interdose anxiety
- Because anxiety disorders can require higher doses, the risk of dependence may be greater in these patients
- Frequency of dosing in practice is often greater than predicted from half-life, as duration of biological activity is often shorter than pharmacokinetic terminal half-life

Overdose

- Fatalities can occur; hypotension, tiredness, ataxia, confusion, coma

Long-Term Use

- Evidence of efficacy for up to 16 weeks
- Risk of dependence, particularly for periods longer than 12 weeks and especially in patients with past or current polysubstance abuse

Habit Forming

- Clorazepate is a Schedule IV drug
- Patients may develop dependence and/or tolerance with long-term use

How to Stop

- Patients with history of seizure may seize upon withdrawal, especially if withdrawal is abrupt
- Taper by 7.5 mg every 3 days to reduce chances of withdrawal effects
- For difficult-to-taper cases, consider reducing dose much more slowly after reaching 30 mg/day, perhaps by as little as 3.75 mg per week or less
- For other patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and